

The quantum speed limit of optimal controlled phasegates for trapped neutral atoms

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Abstract

We study controlled phasegates for trapped neutral atoms in an optical potential. A trapped atom polarizes the vacuum, therefore the ground and excited states of the atom-vacuum system are subject to a Jahn-Teller $1/X$ interaction. We fully account for this interaction in the optimal control theory to calculate the pulse shapes. This allows us to determine the minimum pulse duration, respectively, gate time if there is required to achieve high fidelity. We accordingly analyze the speed-limiting factors, that we find are gate time or fidelity, rather than the interaction strength in the excited state or by the ground state vacuum excitation or the trap. The latter result is the result of the analysis in order to fully account for the rotational state of the atoms at the end of the gate.

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1. Introduction

The physical realization of a quantum computer requires the implementation of a universal gate set [1]. The more difficult goal is providing the necessary gate set, a interaction, interaction between two qubits and a target quantum operation. In principle, for quantum computing with trapped neutral-atom lattices [2–4], the two-qubit gate operation requires atoms near neighbours of distance [5, 6] interaction following adiabatic processes. This implies temperatures much lower than those characteristic of the trap, typically around a few nK of 100 nK. With two-qubit interactions, the dipole-dipole forces between Rydberg atoms [5–8] are employed, the external energy scales are larger and gate speed can in principle reach a few GHz.

Here, we study the limits in the two-qubit gate operation due to the increased interaction of two adjacent atoms that is dominantly excited-molecular ions. This may be a limiting step, like those and the photoionizing two atoms to ions is inevitable [9–11] or high-fidelity Rydberg states [6, 8]. In this context, the system dynamics has to be modelled, including lattice under the influence of the external

laser potential [1]. A high-fidelity gate can then be kept by keeping the band [1]. Fortunately, most any gate operation corresponds to a unitary transformation on the qubit basis, the implementation can be formulated as a minimum control problem [11, 12]. Solutions to this control problem can be found in this framework of optimal control [13, 14]. Such an approach has been explored theoretically for molecular quantum computing with qubits encoded in vibrational states [15, 16]. Especially, the control problem is solved using time-optimal gate pulse shaping combined with feedback loops [17]. Implementation of single-qubit gates with coupled resonant pulses has recently been demonstrated for a system encoded in Rydberg states of an atomic ion [18]. In this experiment, the fundamental limit of the gate operation is set by the success of the Hamiltonian solving since the trajectory dynamics is required to realize arbitrary gate-operations. The gate duration is in particular much shorter than the period of atomic motion in the trap [14].

The goal in this manuscript is, controlling phasegates, implement the 1 -qubit gates, 00 , 01 , 10 and 11 as

$$\hat{U} = \exp(i\theta \hat{\sigma}_1 \hat{\sigma}_2) \quad (1)$$